

RUIZHANG YANG

KAUST Global Fellow

King Abdullah University of Science and Technology (KAUST) | yangrz@eee.hku.hk

RESEARCH PROFILE

Power and energy systems researcher focusing on the planning, operation, and resilience of emerging energy infrastructures, with particular interests in AI data centers, electrified transportation, demand-side flexibility, and data-driven decision-making under uncertainty. Research methods span generative AI and diffusion models, distributionally robust optimization, decision-dependent uncertainty, risk-aware optimization, and digital twins. First- or corresponding-author of 14 published or accepted papers, including 8 Q1 journal papers, with 6 papers published in *IEEE Transactions on Smart Grid* and *Applied Energy*. Four granted invention patents and prior engineering experience in railway power-supply infrastructure.

FELLOWSHIP & RESEARCH LEADERSHIP

KAUST Global Fellowship (2026-2029)

Selected through a global competitive process (approximately 1:30 selection ratio) with approximately USD 285,000 in research support. Leads an independent research project on net-zero and resilient grid-interactive infrastructure corridors for Saudi Vision 2030, integrating energy, transportation, and critical-infrastructure digital twins with optimization-based decision frameworks.

ACADEMIC & PROFESSIONAL APPOINTMENTS

King Abdullah University of Science and Technology (KAUST) | Thuwal, Saudi Arabia Jul. 2026-Present
KAUST Global Fellow, Computer, Electrical and Mathematical Sciences and Engineering Division

- Host: Prof. Charalambos Konstantinou.
- Research on coordinated planning and operation of AI data centers, transportation energy systems, and power grids.

China Railway Siyuan Survey and Design Group Co., Ltd. (CRSCD) | China Jul. 2020-Aug. 2022
Engineer - Transportation Energy Infrastructure Planning, Design, and Engineering Implementation

EDUCATION

The University of Hong Kong | Hong Kong Sep. 2022-Jul. 2026
Ph.D. in Electrical and Electronic Engineering

- Advisor: Prof. Yunhe Hou (IEEE Fellow)
- Research: Optimization and operation of modern power and complex energy systems; machine-learning-driven energy modeling and decision-making.

Huazhong University of Science and Technology | Wuhan, China Sep. 2017-Jul. 2020
M.Eng. in Electrical Engineering

- Advisor: Prof. Jinyu Wen
- Research: High-voltage direct-current transmission systems.

Huazhong University of Science and Technology | Wuhan, China Sep. 2013-Jul. 2017
B.Eng. in Electrical Engineering | GPA: 90.50/100

- Advisor: Prof. Jinyu Wen

RESEARCH INTERESTS

- Grid-interactive emerging energy infrastructures: coordinated planning and operation of AI data centers, transportation energy systems, desalination, buildings, and energy communities, with emphasis on grid access, capacity expansion, demand-side flexibility, low-carbon operation, and resilience to extreme events.
- Data-driven and uncertainty-aware energy decision-making: diffusion models and generative AI, distributionally robust optimization, decision-dependent uncertainty, and risk-aware optimization for system planning, operations, control, scenario modeling, and demand response.

JOURNAL PUBLICATIONS

*Published or accepted. The author name is shown in bold; * denotes corresponding author.*

- J1. **R. Yang**, Y. Hou "Grid-Interactive and Dual-Side Demand Responsive Timetable Scheduling Optimization for Flexible Urban Metro Systems," *IEEE Transactions on Smart Grid*, 2026, doi: 10.1109/TSG.2026.3660334.
- J2. **R. Yang**, Y. Li, Z. Xiao, and Y. Hou "Coordinative Operating Strategy of Grid-Interactive FTPSS-HST for Cost-Efficient Operation," *IEEE Transactions on Smart Grid*, vol. 16, no. 1, pp. 146-163, Jan. 2025.
- J3. **R. Yang**, Y. Hou "Resilience Enhancement of Railway Energy System: A Multi-Stage Strategy Integrating FTPSS Configuration, Emergency Energy Management and Adaptive Train Control," *Applied Energy*, vol. 402, 126910, 2026.
- J4. **R. Yang**, C. Konstantinou, and Y. Hou "Enabling Grid-Interactive Data Center-Desalination Coordination through Thermo-Electric Coupling-Based Waste Heat Utilization," *Applied Energy*, vol. 408, 127428, 2026.
- J5. **R. Yang**, C. Konstantinou, and Y. Hou "Leveraging Heterogeneous Railway Systems as Virtual Inertia Sources for Grid Frequency Regulation," *IEEE Transactions on Smart Grid*, early access, 2026, doi: 10.1109/TSG.2026.3706385.
- J6. **R. Yang**, Y. Li, C. Hu, and F. Y. Hou "ExDiffusion: Classifier-Guidance Diffusion Model for Extreme Load Scenario Generation With Extreme Value Theory," *IEEE Transactions on Smart Grid*, vol. 16, no. 5, pp. 3887-3903, Sep. 2025.

- J7. W. Xiang, **R. Yang***, C. Lin, J. Zhou, J. Wen, and W. Lin “A Cascaded Converter Interfacing Long-Distance HVdc and Back-to-Back HVdc Systems,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 8, no. 4, pp. 4109-4121, Dec. 2020.
- J8. **R. Yang**, B. Li, J. Yang, et al. “An LCC-MMC Hybrid Cascaded Inverter Applicable for UHVDC Power Overhead Line Transmission and Dynamic Reactive Power Self-Compensation,” *Energy Reports*, vol. 6, pp. 943-952, 2020.
- J9. R. Cui, C. Hu, **R. Yang**, and F. Y. Hou “Co-Optimization of Coincident Peak Management and Ancillary Services for Flexible Data Centers,” *IEEE Transactions on Smart Grid*, early access, 2026, doi: 10.1109/TSG.2026.3698492.
- J10. **R. Yang**, W. Xiang, H. Rao, et al. “A Cascaded Converter Valve for High-Capacity Power Infeed and Asynchronous Grid Interconnection,” *Proceedings of the CSEE*, vol. 39, no. 17, pp. 5182-5194, 5299, 2019. [in Chinese]
- J11. **R. Yang**, W. Xiang, W. Li, and J. Wen “Hybrid Ultra-HVDC System with LCC and Cascaded Hybrid MMC,” *The Journal of Engineering*, no. 16, pp. 1112-1116, 2019.

MANUSCRIPTS UNDER REVIEW OR REVISION

- M1. **R. Yang**, C. Konstantinou, and Y. Hou “Node-Aware Spatiotemporal Carbon Tracing and Carbon-Responsive Control of Electrified Railways for Emission Reduction and Tertiary Reserves,” *Applied Energy*, minor revision.
- M2. **R. Yang**, C. Konstantinou, and F. Y. Hou “Uncertainty-Aware Online Demand Response Baseline Estimation via Causal Diffusion Models for Resilient Demand-Side Management,” *Applied Energy*, major revision.
- M3. **R. Yang**, C. Konstantinou, and Y. Hou “Multi-Stage Cross-Regional Data-Center Portfolio Expansion under Grid-Interconnection Constraints and Decision-Dependent Workload Uncertainty,” *IEEE Transactions on Power Systems*, manuscript under review.
- M4. **R. Yang**, C. Konstantinou “Distributionally Robust Capacity Contracting for Heterogeneous AI Colocation under Power-Security Constraints,” *IEEE Transactions on Network Science and Engineering*, manuscript under review.
- M5. **R. Yang**, Z. Xiao, and C. Konstantinou “A Multi-Timescale Closed-Loop Digital-Twin Framework for Resilient FTPSS-Coupled HST Operation under Extreme Weather,” *IEEE Transactions on Transportation Electrification*, manuscript under review.
- M6. **R. Yang**, Y. Wang, F. Y. Hou, and C. Konstantinou “Human Mobility-Driven and Network-Aware Energy Sharing among Heterogeneous Communities: A Distributionally Robust Game Approach,” *IEEE Transactions on Smart Grid*, manuscript under review.

CONFERENCE PAPERS

- C1. **R. Yang**, M. Zhang, and Y. Hou “Coordinative Operation for Grid-Interactive Efficient Buildings Considering Thermal Dynamic,” *IEEE PES ISGT Europe*, pp. 1-6, Oct. 2024, Croatia.
- C2. **R. Yang**, W. Xiong, and Y. Hou “Probabilistic Lifecycle Costing Evaluation for Configuration of Transformers in High-Speed Railway Based on Optimal KDE,” *IEEE PES General Meeting*, pp. 1-6, Jul. 2023, USA.
- C3. **R. Yang**, W. Xiang, W. Li, et al. “A Hybrid Cascaded Converter Applicable for UHVDC Transmission and Feeding to Load Centre,” *15th IET International Conference on AC and DC Power Transmission (ACDC 2019)*, 2019.

HONORS & TALENT PROGRAMS

KAUST Global Fellowship (2026-2029)

- Global competitive selection with an approximate 1:30 selection ratio.
- Approximately USD 285,000 in research support for an independent fellowship project.

RESEARCH PROJECTS

National Key R&D Program of China - Core team member

Mathematical Models and Optimization Algorithms for Operational Reliability of Large-Scale Complex Power Systems

- Mathematical modeling and optimization for operational reliability of complex power systems.
- Security and risk-aware decision-making under high-dimensional uncertainty.

National Natural Science Foundation of China Joint Fund - Core team member

Theory and Methods for Supply-Demand Balancing in Power Systems with High Penetrations of Renewable Energy

- Supply-demand balancing mechanisms under high renewable uncertainty.
- Flexibility resource allocation and coordinated regulation for renewable-rich power systems.

National Key R&D Program of China - Project participant

Flexible HVDC Transmission Technology for UHV, High-Capacity, Long-Distance Overhead Lines

- Converter topology and system-control research for flexible HVDC transmission systems.
- Modeling, analysis, and technical validation of UHV high-capacity transmission equipment.

TEACHING & ACADEMIC SERVICE

- Teaching Assistant for undergraduate and professional courses including Electricity and Electronics and Power System Analysis and Control at The University of Hong Kong; approximately 100 teaching-assistant hours per semester.
- Recipient of a higher-education teaching qualification/certificate from The University of Hong Kong.
- Reviewer for IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, Applied Energy, and other academic journals.